

Sequence numbers SHALL be separately defined for either direction of a BTP session. The sequence number of the first packet sent by the client after completion of the BTP session handshake SHALL be zero. The server's BTP handshake response bears an implied sequence number of zero because it occupies a slot in the client's receive window. The client acknowledges the server's BTP handshake response with an acknowledgement sequence of zero. For this reason, the sequence number of the first data packet sent by the server after completion of the BTP session handshake SHALL be 1.

Peers SHALL check to ensure that all received BTP packets properly increment the sender's previous sequence number by 1. If this check fails, the peer SHALL close the BTP session and report an error to the application.

4.19.4.7. Receive Windows

The purpose of the receive window is to enable flow control at the GATT session layer between BTP peers.

Flow control is required at the GATT transport layer for embedded platforms that use "minimal" BLE chipsets. These platforms may have limited space on the host processor to receive packets from their BLE chipsets. In the case of some dual-chip architectures, writes and indications are received and confirmed by the BLE chip with no input from the host processor. When the BLE chip sends the result of a received GATT PDU to the host processor, that payload and the corresponding BTP packet will be permanently lost if the host does not have enough space to receive it. For this reason, knowledge of a remote host's ability to reliably receive GATT PDUs is presented at the transport layer in the form of the BTP receive window.

Both peers in a BTP session SHALL define a receive window, where the window's size indicates the number of GATT PDUs (that is, BTP segments) a peer can reliably receive and store without session-layer acknowledgment. A maximum window size SHALL be established for both peers as part of the BTP session handshake. To prevent sequence number wrap-around, the largest maximum window size any peer may support is 255.

Both peers SHALL maintain a counter to reflect the current size of the remote peer's receive window. Each peer SHALL decrement this counter when it sends a packet via GATT write or indication and increment this counter when a sent packet is acknowledged.

If a local peer's counter for a remote peer's receive window is zero, the window SHALL be considered closed, and the local peer SHALL NOT send packets until the window reopens (is incremented above zero). When a closed window reopens, a local peer SHALL immediately resume any pending BTP packet transmission.

A local peer SHALL also not send packets if the remote peer's receive window has one slot open and the local peer does not have a pending packet acknowledgement. This is to avoid the situation where the receive windows of both peers are full and neither can send an acknowledgement to reopen its window for the other. Because the server's handshake response bears an implicit BTP sequence number of zero, a server SHALL initialize its counter for the client's receive window size at (negotiated maximum window size - 1). A client SHALL initialize its counter for the server's receive window at the negotiated maximum window size.

Both peers SHALL also keep a counter of their own receive window size based on the sequence